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# Autocatalytic Chemiluminescence Sheds New Light on the Classic Permanganate–Oxalate Reaction

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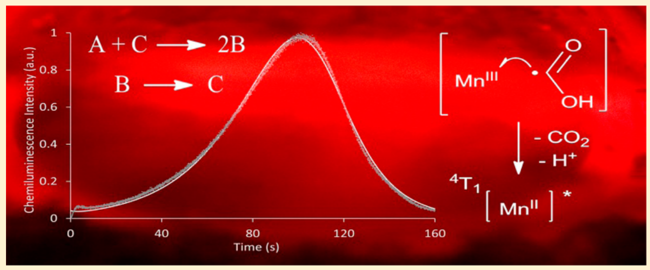
Supporting Information

**ABSTRACT:** The emission of light from the permanganate–oxalate reaction enables monitoring of intermediates not accessible through traditional spectrophotometric interrogation. Despite the inherent complexity of the underlying chemical reactions and equilibria, the emission intensity–time profile was characterized by a simple model combining previously independent minimalistic descriptions of chemiluminescence and autocatalysis. The generation of the electronically excited $[Mn^{II}]^*$ emitter and the acceleration of the reaction even in the presence of high initial concentrations of $Mn^{II}$ (under conditions that preclude accumulation of colloidal $Mn^{IV}$) provide new evidence for the reduction of manganese species by a reactive radical intermediate as a supplementary positive feedback loop to the formation of $Mn^{II}$.






## 1. INTRODUCTION

The reaction between oxalate and permanganate (eq 1) is a classic redox system that for many decades has served as an undergraduate laboratory exercise in titrimetric analysis and a textbook example of autocatalytic reaction kinetics.

$$2MnO_4^{-} + 5C_2O_4^{2-} + 16H^{+} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O \quad (1)$$

The study of this system began in the 19th century,[1] and in spite of extensive investigation (almost exclusively involving UV–visible spectrophotometry),[2–9] the reaction pathway is yet to be fully elucidated. It is generally accepted, however, that the permanganate is primarily reduced by a $[Mn^{II}(C_2O_4)]$ complex (eq 3),[4,8,9] which is initially present due to $Mn^{II}$ impurities.[2,5] Manganic ions generated via eqs 3 and 4 form $[Mn^{III}(C_2O_4)_n]^{(3-2n)}$ complexes that decompose to regenerate the $Mn^{II}$ autocatalyst (eq 6; where $n$ = 1, 2, or 3 depending on the solution acidity and oxalate–manganese ratio).[3,4]

$$Mn^{2+} + C_2O_4^{2-} \rightarrow [Mn(C_2O_4)] \quad (2)$$

$$MnO_4^{-} + [Mn(C_2O_4)] + 4H^{+} \rightarrow MnO_2 + Mn^{3+} + 2CO_2 + 2H_2O \quad (3)$$

$$[MnO_2, H_2C_2O_4] + 2H^{+} \rightarrow Mn^{3+} + CO_2 + CO_2^{\bullet -} + 2H_2O \quad (4)$$

$$Mn^{3+} + nC_2O_4^{2-} \rightarrow [Mn(C_2O_4)_n]^{(3-2n)} \quad (5)$$

$$[Mn(C_2O_4)_n]^{(3-2n)} \rightarrow Mn^{2+} + CO_2 + CO_2^{\bullet -} + (n-1)C_2O_4^{2-} \quad (6)$$

$$2CO_2^{\bullet -} \rightarrow C_2O_4^{2-} \quad (7)$$

Pimienta et al.[8] sought a complete description of the reaction and proposed a 14 step model (including 8 equilibria). The model simulated some important characteristics of the reaction, but only when incorporating the direct reduction of permanganate by oxalic acid, which contradicted several earlier observations.[2,9] Kovács et al. subsequently demonstrated that the reduction of $Mn^{III}$ cannot solely account for the observed acceleration of eq 3 and that the reaction exhibits positive feedback even if a large excess of $Mn^{II}$ is present.[9] These findings suggest a second autocatalytic agent, which they tentatively ascribed to colloidal $Mn^{IV}$, but noted an alternative explanation in which the radical ions produced in eq 6 react with permanganate.

Our interest in this matter stems from observations that the reaction is chemiluminescent.[10–12] The emission was previously ascribed to side reactions involving the formation of molecular oxygen[10] or a carbon dioxide dimer[11] and ignored in

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subsequent investigations into the reaction mechanism.[5−9] Herein we show that the light-producing pathway is inextricably linked to the production and consumption of intermediates with key roles in the characteristic acceleration of the reaction, which enables investigation of intermediates that are not accessible through traditional spectrophotometric methods.

## 2. EXPERIMENTAL SECTION

Absorption spectra were acquired using a Cary 300 UV−vis spectrophotometer (Agilent) with quartz cuvettes (1 cm path length), using a scan rate of 600 nm $min^{-1}$ and a bandwidth of 1 nm. Chemiluminescence spectra were acquired using a Cary Eclipse fluorescence spectrophotometer (Agilent) with R928 photomultiplier tube, in 'Bio/chemiluminescence' mode (gate time, 1000 ms; data interval, 5 nm; slit, 20 nm; PMT voltage, 800 V). Reactant solutions were merged at a T-piece prior to entering a coiled flow-cell, which was mounted against the emission window of the spectrometer. Final spectra are an average of 50 scans. The correction factor was established as previously described.[13]

Stopped-flow experiments were performed with a manifold consisting of a programmable dual syringe pump (Model sp210iw, World Precision Instruments, Glen Waverly, Victoria, Australia), Valco six-port injection valve (SGE, Ringwood, Victoria, Australia), and 'GloCel' chemiluminescence detector with dual-inlet serpentine-channel reaction zone[14] (Global FIA, Fox Island, WA, USA). In this dual-inlet configuration, the solutions were merged in front of the photomultiplier module (Electron Tubes model P30A-05; ETP, Ermington, NSW, Australia), and therefore, the entire chemiluminescence intensity versus time profile was captured. The syringes were loaded with 2 M sulfuric acid (carrier line) and the potassium−oxalate solution, also in 2 M sulfuric acid. The carrier line was connected to a 6-port (2-position) valve, with a 70 $\mu$L injection loop that was filled with the permanganate solution. When the pump was activated, 120 $\mu$L of the carrier and oxalate solutions were dispensed from the syringes (at a flow rate of 10 mL $min^{-1}$ per line). This propelled the permanganate and oxalate solutions into the reaction channel (within 0.8 s), where the mixture was held for a set period of time. The output signal from the photomultiplier module was recorded via an e-corder 410 data acquisition module (eDAQ, Denistone East, NSW, Australia), using Chart v5 software (eDAQ), with 20 measurements per second. Longitudinal dispersion of the permanganate solution zone was minimized using the shortest possible length of tubing between the valve and detector. Inserting a small volume of the permanganate solution into a carrier stream in this manner, rather than dispensing a permanganate stream from the syringe pump, enabled thorough flushing of the detector. The pump was activated for an extended period of time (without filling the injection loop with permanganate) after each profile was collected, to remove the $Mn^{II}$ product from the detection flow cell.

The experimental data were fitted with kinetic models in which the parameters (initial concentrations of species and rate coefficients) were optimized. The differential rate laws were numerically integrated using a first order Euler method.[15,16] The parameters were varied using a simplex algorithm implemented in the Solver feature in Microsoft Excel, until the sum of squares of the difference between the experimental and fitted data was minimized.

## 3. RESULTS AND DISCUSSION

**3.1. Nature of Manganese Intermediates.** Considering Pimienta and co-workers' observation[17] that the proportion of the $Mn^{IV}$ intermediate decreased with increasing $H_2SO_4$ concentration, we have examined the formation of $Mn^{IV}$ over a wide range of acid concentrations through spectrophotometric titrations of permanganate with thiosulfate (Figures 1 and S1, Supporting Information). With $H_2SO_4$ concentrations below 1 M, the reduction resulted in broad bands across the visible region, previously attributed to colloidal $Mn^{IV}$.[18] In near-neutral solution, the colloid was stable ($\lambda_{max} \approx 360$ nm), and no further change in absorbance was observed upon the addition of thiosulfate beyond the stoichiometric ratio. Flocculation of $Mn^{IV}$ was observed in $H_2SO_4$ between 0.1 mM and 100 mM. However, in 2 M $H_2SO_4$ (Figure 1b), the characteristic bands of permanganate decreased to leave a weak band at ∼485 nm attributable to $Mn^{III}$,[19] with no evidence of $Mn^{IV}$ during the time frame of the experiment.

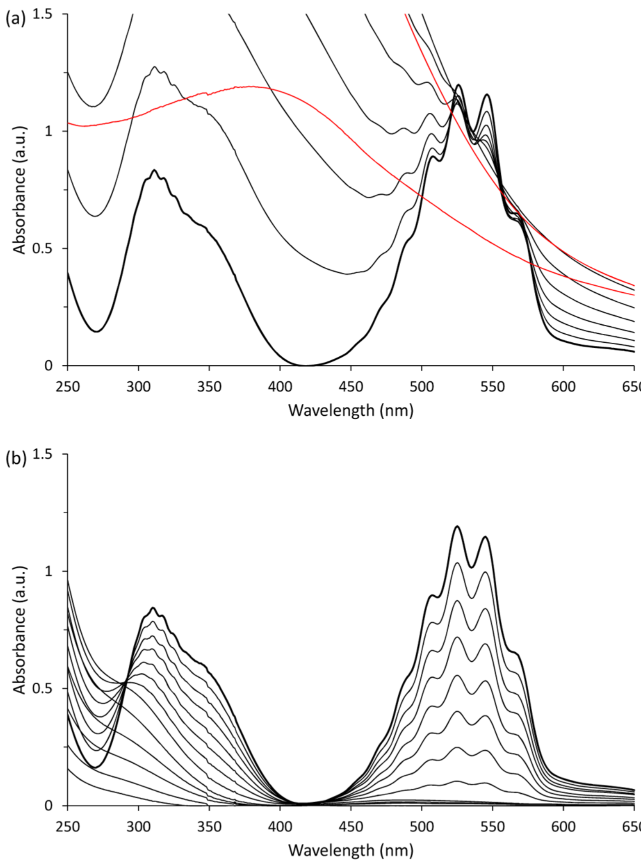



**Figure 1.** UV−vis titration of (a) 2 L of $KMnO_4$ (0.5 mM in 0.1 mM $H_2SO_4$) with 8 × 4 mL aliquots of $Na_2S_2O_3$ (18.8 mM). Red lines, spectra recorded after the addition of the final two aliquots; and (b) 2 L of $KMnO_4$ (0.5 mM in 2 M $H_2SO_4$) with 12 × 4 mL aliquots of $Na_2S_2O_3$ (18.8 mM).

The reaction of permanganate and oxalate has often been studied with $H_2SO_4$ concentrations between 5 and 100 mM,[5,7,8,17] which promote a higher ratio of $Mn^{III}/Mn^{IV}$ than that observed in less acidic solutions (Figure S1, Supporting Information).[20] In contrast, Kovács et al.[9] did not add $H_2SO_4$ or $HClO_4$ to their reaction mixture, although they did use oxalic acid as their source of oxalate. Our experiments were conducted under conditions that, unlike those of Kovács et al., avoid the accumulation of $Mn^{IV}$.

**3.2. Light Producing Pathway.** We re-evaluated the previous proposals regarding the light-emitting species,[10,11] considering the findings of more recent investigations of other chemiluminescent permanganate reactions: (1) a broad red emission has been observed with a wide range of organic substrates[21] (although some exceptions are known[22]); and (2) the characteristic red emission ($\lambda_{max}$ = 734 ± 5 nm) from these reactions has been identified as the $^4T_1 \rightarrow {}^6A_1$ transition of $Mn^{II}$.[23]

Obtaining an emission spectrum from this system is hindered by the slow rate of reaction, but by continuously merging oxalate with permanganate solutions after a preliminary partial reduction of the oxidant with sodium thiosulfate, we observed the characteristic emission of $[Mn^{II}]^*$ (Figure S2, Supporting Information).

Using electron paramagnetic resonance (EPR) spectroscopy, we have recently shown that the $[Mn^{II}]^*$ emitter in reactions between permanganate and organic substrates is generated through one-electron reduction of $Mn^{III}$ by the radical intermediates of substrate oxidation.[24] The formation of intermediates ($C_2O_4^{\bullet-}$ and/or $CO_2^{\bullet-}$) in the title reaction is widely acknowledged,[3,7−9] but their fate is the source of much conjecture. Numerous researchers have proposed the reduction of manganese species ($Mn^{III}$ and/or $Mn^{IV}$) by radical intermediates.[4,25] Kinetic analysis with radical scavengers have suggested that such reactions do occur,[7] but in recent discussions,[8,26] these pathways have been overlooked. Similarly, Perez-Benito et al.[27] reported autocatalysis in the reduction of oxalic acid by colloidal $MnO_2$ (formed in neutral solution), which they attributed to a pathway involving adsorption of both oxalic acid and $[Mn^{II}(C_2O_4)]$ on the particle surface. However, they also commented that the acceleration may arise from the reduction of $MnO_2$ by $CO_2^{\bullet-}$. Our confirmation that the chemiluminescence in the permanganate−oxalate reaction emanates from $[Mn^{II}]^*$ provides new support for the participation of these reactive intermediates in the reduction of higher Mn oxidation states.[24] The exact species involved in chemiexcitation is not known, but of the two candidates, we favor $CO_2^{\bullet-}$ over $C_2O_4^{\bullet-}$, bearing in mind the free energy of the fragmentation reaction has been estimated at −0.61 eV (and $K = 3 \times 10^{10}$ $M^{-1}$).[28] In the case of the mono-oxalato $Mn^{III}$ complex, the light producing pathway can be derived from Taube's description[3] of its decomposition:

$$[Mn^{III}(C_2O_4)]^+ \rightarrow Mn^{2+} + C_2O_4^{\bullet-} \tag{8}$$

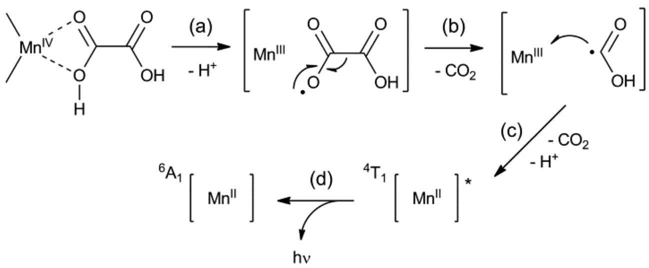
$$C_2O_4^{\bullet-} \rightarrow CO_2^{\bullet-} + CO_2 \tag{9}$$

$$[Mn^{III}(C_2O_4)]^+ + CO_2^{\bullet-} \rightarrow [Mn^{II}(C_2O_4)]^* + CO_2 \tag{10}$$

$$[Mn^{II}(C_2O_4)]^* \rightarrow [Mn^{II}(C_2O_4)] + h\nu \tag{11}$$

This pathway is analogous to the classic coreactant electrochemiluminescence mechanism for oxalate and $[Ru^{II}(bpy)_3]^{2+}$, first postulated by Rubinstein and Bard,[29] and recently confirmed in our laboratories through direct EPR characterization of the protonated radicals captured on a quartz surface.[30] The $pK_a$ of the $CO_2^{\bullet-}/HCO_2^{\bullet}$ couple is 2.3,[31] so the protonated form will also dominate under the conditions of the permanganate−oxalate reaction. This pathway can be extended to the light-producing[12] reaction (or reductive dissolution[32]) of $Mn^{IV}$ species with oxalate (eq 4) to form $Mn^{III}$ and the radical in close proximity, increasing the likelihood of their reaction (Scheme 1).

**Scheme 1. Light-Producing Pathway for the Reductive Dissolution of $Mn^{IV}$ with Oxalate[a]**




[a](a) Electron transfer from an adsorbed reducing agent to the metal center;[32] (b) fragmentation of the initial radical intermediate;[28] (c) reduction of $Mn^{III}$ to generate $[Mn^{II}]^*$;[24] (d) the characteristic red emission.[12,23]

**3.3. Chemiluminescence Profiles.** We examined the rate of the light-producing reaction between oxalate and $Mn^{III}$ in isolation by slowly adding permanganate to $MnSO_4$ in 2 M $H_2SO_4$, before using stopped-flow methodology to combine $Mn^{III}$ with oxalate in 2 M $H_2SO_4$. The intensity-time profile for this reaction returned to baseline within 20 s (Figure S3, Supporting Information). In contrast, the emission profile for the reaction of oxalate and permanganate (Figure 2; red trace) displayed a significant induction period and persisted for over 160 s.

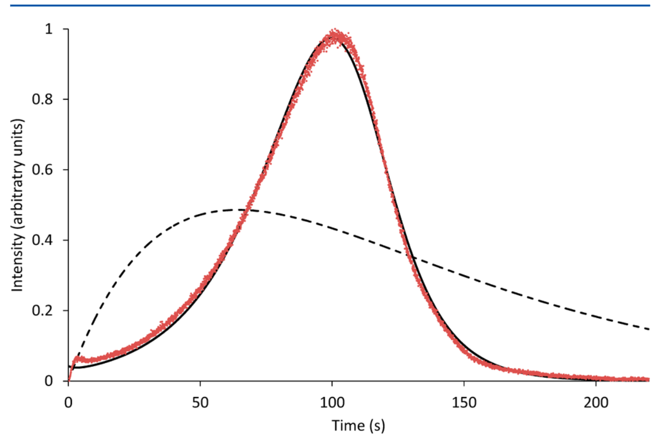



**Figure 2.** Experimental chemiluminescence intensity−time profile (red trace) for the reaction between permanganate (0.5 mM) and oxalate (1 mM) in acidic aqueous solution, and the fitted double-exponential (dashed black trace) and autocatalytic (solid black trace) models.

To explore the inherent kinetic information of this transient emission, we sought a simple model akin to the double-exponential description of chemiluminescence,[33,34] which is based on two pseudo first-order reactions (eq 12), where A, B, and C represent pools of reactants, intermediates, and products, respectively, and chemiluminescence intensity is proportional to pool B.

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \tag{12}$$

This conventional model provided an excellent fit for the emission accompanying the reaction of oxalate and $Mn^{III}$

(Figure S3, Supporting Information), but a very poor fit for the profiles generated from the reaction of oxalate and permanganate (Figure 2, dashed black trace). Drawing from the differential rate law adopted by Perez-Benito et al. to characterize the autocatalytic reduction of permanganate by various compounds (eq 13)[35] and the minimalistic description for autocatalysis[36] championed by Finke and co-workers (eq 14),[37,38] we examined several possible models of the chemiluminescence profile that incorporated autocatalytic steps.

$$r = k_1[R] + k_2[R][P] \tag{13}$$

$$\mathrm{A} \xrightarrow{k_1} \mathrm{B}$$
$$\mathrm{A} + \mathrm{B} \xrightarrow{k_2} 2\mathrm{B} \tag{14}$$

Of those examined, the best fit (Figure 2, solid black trace) was obtained using the following model:

$$\mathrm{A} \xrightarrow{k_1} \mathrm{B}$$
$$\mathrm{B} \xrightarrow{k_2} \mathrm{C}$$
$$\mathrm{A} + \mathrm{C} \xrightarrow{k_3} 2\mathrm{B} \tag{15}$$

$$\frac{\mathrm{dA}}{\mathrm{d}t} = -k_1[\mathrm{A}] - k_3[\mathrm{A}][\mathrm{C}]$$
$$\frac{\mathrm{dB}}{\mathrm{d}t} = k_1[\mathrm{A}] + 2k_3[\mathrm{A}][\mathrm{C}] - k_2[\mathrm{B}]$$
$$\frac{\mathrm{dC}}{\mathrm{d}t} = k_2[\mathrm{B}] - k_3[\mathrm{A}][\mathrm{C}] \tag{16}$$

While we are cautious about assigning specific aspects of the oxalate−permanganate reaction to each step of this minimalistic empirical description, the autocatalytic pathway of this model is indeed reminiscent of eq 3. Remarkably, the fitting process gave a $k_1$ value of zero, and therefore, in spite of the complexity of the underlying reactions and equilibria,[8,9] the chemiluminescence profile was well characterized by only two steps: A + C → 2B → 2C. The absence of an A → B step necessitates an initial value for B or C. These observations support previous claims that the reaction does not proceed (at least not at a significant rate) in the absence of the autocatalyst, which is initially present as an impurity.[2,9] As described above, the manganese impurity will be present in the permanganate solution in a higher oxidation state than $Mn^{II}$. The initial, rapid reaction of this intermediate manganese species with oxalate can be observed as a small sharp rise in intensity at the beginning of the chemiluminescence profile ($t$ = 0 to 3 s, Figure 2).

Demonstrating the broad applicability of the approach, we also successfully applied the model to the previously published[39] chemiluminescence profiles for the autocatalytic oxidation of lucigenin in the presence of platinum nanoparticles in a mixed water−ethanol solvent (Figure 3), which exhibit much longer induction periods (up to 350 s). Again, the best fit was obtained with a $k_1$ value of zero, and only two steps (A + C → 2B → 2C) were required.

### 3.4. Addition of $Mn^{II}$.

Considering the complexity of the reaction, the above model has simplified many aspects, and the question thus arises: Does the single autocatalytic step account for more than one positive feedback loop? To address this

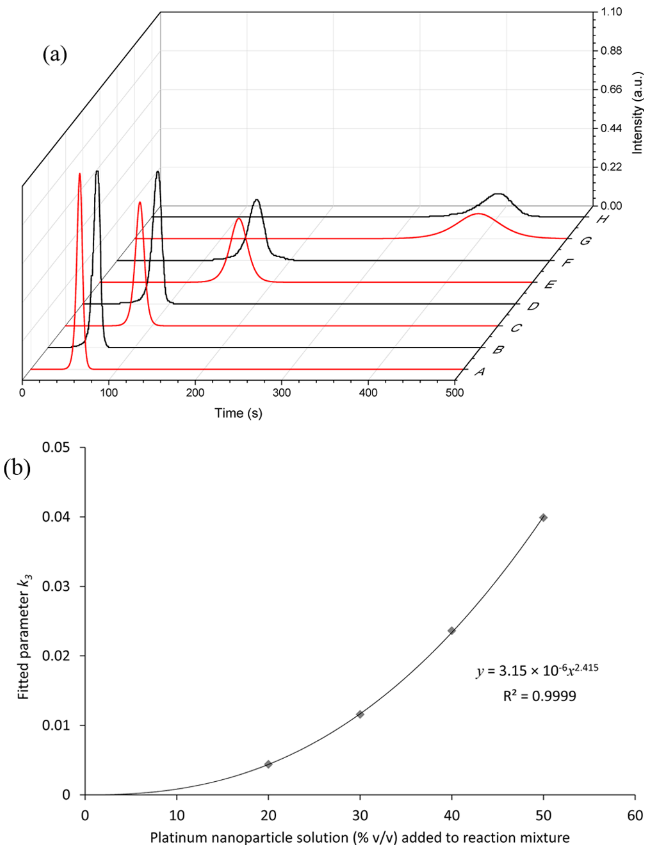



**Figure 3.** (a) Chemiluminescence intensity versus time profiles for the autocatalytic oxidation of lucigenin initiated by platinum nanoparticles in a mixed water−ethanol solvent. Red curves: digitized data[39] representing the profiles obtained with (A) 50%, (C) 40%, (E) 30%, and (G) 20% v/v addition of the platinum nanoparticle solution to the total reaction mixture. The data has been normalized with respect to curve A. Black curves: characterization of the data using our minimalistic model of autocatalytic chemiluminescence, where $k_3$ is (B) 0.0399, (D) 0.0236, (F) 0.0116, and (H) 0.00439. Other parameters (held consistent for all four profiles): $k_1 = 0$, $k_2 = 2.95$, $[\mathrm{A}]_0 = 13.96$, $[\mathrm{B}]_0 = 0$, and $[\mathrm{C}]_0 = 1 \times 10^{-9}$. In each case, the reaction is initiated at $t = 0$. (b) The relationship between platinum nanoparticle concentration (as stated in ref 39) and fitted parameter $k_3$.

issue, we introduced relatively high concentrations of the $Mn^{II}$ catalyst under conditions that impede the accumulation of $Mn^{IV}$ (Figure 1) and examined its influence on the intensity−time profiles.

The addition of $Mn^{II}$ (0.05 or 0.25 mM) to the oxalate solution reduced the time required to reach maximum intensity (Figure 4a). However, in spite of its presence at levels much higher than those normally encountered before the final stages of the reaction,[9] an induction period was evident in both profiles. In fact, the increase in emission intensity continued to accelerate until 22.4 and 14.4 s (the inflection points in Figure 4a are more clearly seen as maxima in the corresponding first derivative plots; Figure 4b). Application of the simple autocatalytic model under these highly distorted conditions resulted in increases in the effective rate coefficients for all model steps, rather than increases in $[\mathrm{C}]_0$ (Figure 4c).

In spite of the long-held notion that the reduction of permanganate is initiated by a complex of oxalate and a $Mn^{II}$ impurity,[2,4,5] a mixture of these Mn species will generate $Mn^{IV}$/$Mn^{III}$ intermediates,[40] and our conditions favor the formation of $Mn^{III}$ (Figure S1, Supporting Information). The initial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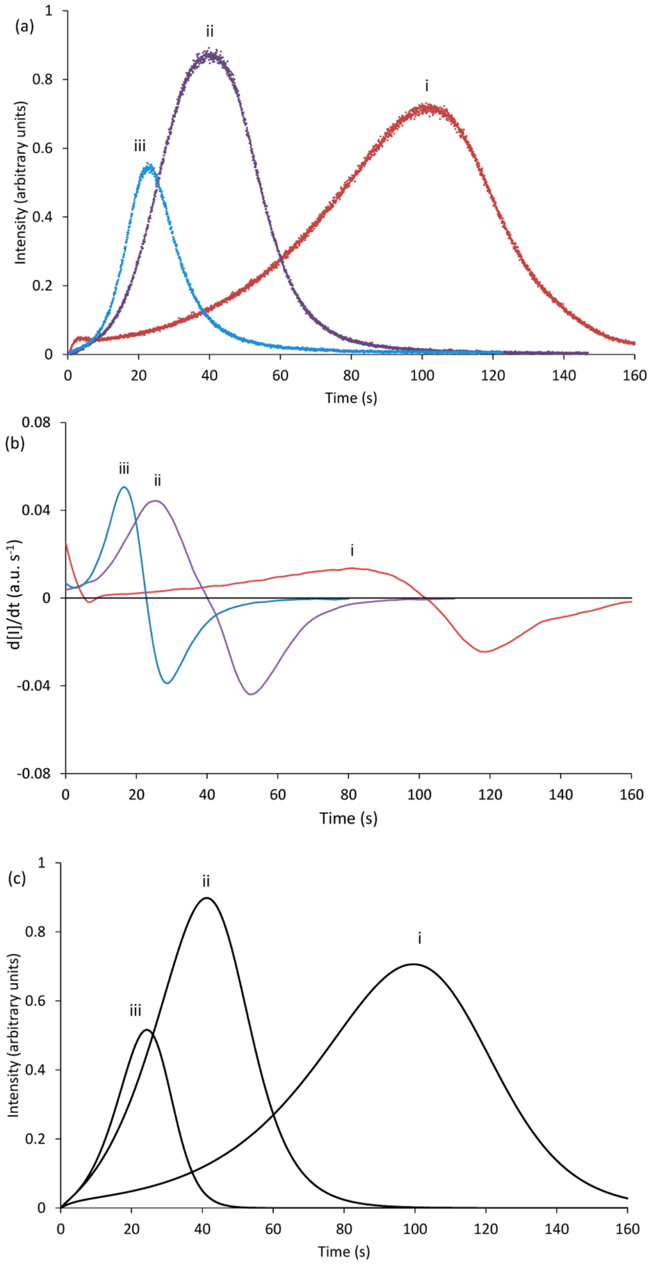


**Figure 4.** (a) Chemiluminescence intensity−time profiles for the reaction between permanganate (0.5 mM in 2 M $H_2SO_4$) and oxalate (1 mM in 2 M $H_2SO_4$) with (i) 0, (ii) 0.05, and (iii) 0.25 mM $Mn^{II}$ added to the oxalate solution. (b) First derivative plots of the experimental chemiluminescence intensity−time profiles. The peak maxima and $x$-intercepts in these plots correspond to the inflection points and maximum intensities of the experimental data. (c) Fitted autocatalytic model. i, $[A]_0 = 2.06$*, $[B]_0 = 0$, $[C]_0 = 0.0222$, $k_1 = 0$, $k_2 = 0.0895$, $k_3 = 0.0695$; ii, $[A]_0 = 2.06$*, $[B]_0 = 0$, $[C]_0 = 0$, $k_1 = 0.00530$, $k_2 = 0.125$, $k_3 = 0.172$; iii, $[A]_0 = 2.06$*, $[B]_0 = 0$, $[C]_0 = 0$, $k_1 = 0.00622$, $k_2 = 0.382$, $k_3 = 0.166$.[41] *Constrained parameter.[42]

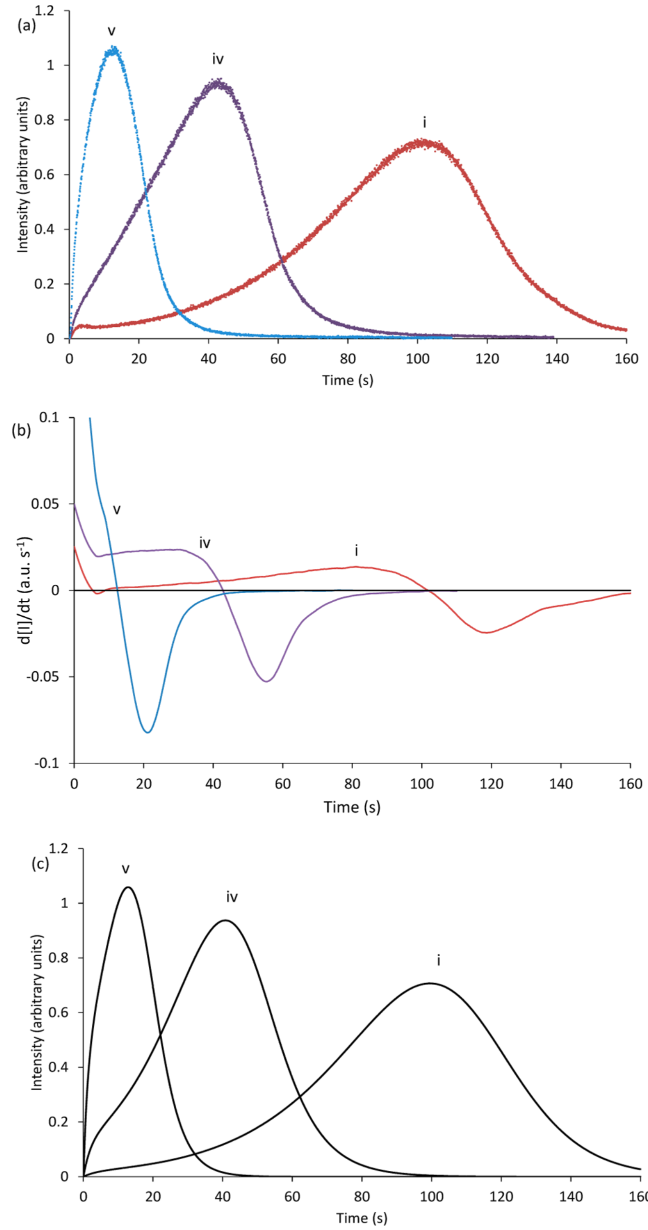


**Figure 5.** (a) Chemiluminescence intensity−time profiles for the reaction between permanganate (0.5 mM in 2 M $H_2SO_4$) and oxalate (1 mM in 2 M $H_2SO_4$) with (i) 0, (iv) 0.05, and (v) 0.25 mM $Mn^{II}$ added to the permanganate solution. (b) First derivative plots of the experimental chemiluminescence intensity-time profiles. (c) Fitted autocatalytic model. Parameters: i, $[A]_0 = 2.13$*, $[B]_0 = 0$, $[C]_0 = 0.0225$, $k_1 = 0$*, $k_2 = 0.0926$, $k_3 = 0.0653$; iv, $[A]_0 = 2.13$*, $[B]_0 = 0$, $[C]_0 = 0.128$, $k_1 = 0$*, $k_2 = 0.114$, $k_3 = 0.129$; v, $[A]_0 = 2.13$*, $[B]_0 = 0$, $[C]_0 = 0.494$, $k_1 = 0$*, $k_2 = 0.208$, $k_3 = 0.190$.[41] *Constrained parameter.[42]

generation of $Mn^{III}$ prior to reaction with oxalate greatly accentuated the sharp initial increase in chemiluminescence intensity (Figure 5), which was not observed when the $Mn^{II}$ was added to the oxalate solution (Figure 4). With greater quantities of this intermediate at the start of the reaction with oxalate, the induction period was effectively removed, where the first derivative plot (Figure 5b) decreased from the initiation of the reaction ($t = 0$) to beyond the $x$-intercept corresponding to the maximum chemiluminescence intensity ($t = 11.5$). Nevertheless, autocatalysis is evident from the good fit using the above model (albeit with variation in multiple model parameters; Figure 5c) and poor fit using the conventional double-exponential model.

## 4. CONCLUSIONS

The light-producing pathway within this reaction is inextricably linked to the production and consumption of intermediates with key roles in the characteristic acceleration of the reaction.

The addition of $Mn^{II}$ to the oxalate solution provides an immediate high concentration of $[Mn^{II}(C_2O_4)]$ to reduce permanganate (eq 3), and the observed acceleration under these conditions was attributed to formation of highly reactive radical intermediates capable of reducing not only $Mn^{III}$ (to generate the emitter), but also higher manganese oxidation states. These findings do not, however, rule out autocatalysis from dissolved or finely dispersed colloidal $Mn^{IV}$ species nor do they contradict the previously demonstrated catalytic effect of colloidal $Mn^{IV}$ in various permanganate reactions under less acidic conditions; rather, they shed new light on the frequently overlooked contribution of radical intermediates in this system. Considering the importance of manganese chemistry, including autocatalytic and oscillatory reactions and environmental systems, we anticipate considerable benefit from wider application of this experimental approach to examine reaction kinetics.

## ■ ASSOCIATED CONTENT

### Ⓢ Supporting Information

Additional discussion on previous experimental approaches versus chemiluminescence; spectrophotometric examination of $Mn^{III}$ and $Mn^{IV}$ formation from reduction of permanganate under acidic conditions; chemiluminescence spectra; chemiluminescence intensity−time profile for the reaction between $Mn^{III}$ and oxalate. This material is available free of charge via the Internet at http://pubs.acs.org.

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### Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

### Notes

The authors declare no competing financial interest.

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(41) In these experiments, the intensity of the emitted light has arbitrary units. The luminescence intensity is proportional to the concentration of the emitting species, but the factor of proportionality is not known (and is dependent on many contributing factors, including the chemiluminescence quantum yield, portion of emitted light captured by the photodetector, and the conversion of light to the voltage output signal). Moreover, the simplified reaction scheme is derived from the classic double-exponential model of chemiluminescence involving generalized pools of reactants, intermediates, and products. Thus, for the purposes of these experiments, the units of A, B, and C are also arbitrary. The units of the rate are a.u. per second, and those of the rate coefficients are a.u. per second, where these latter arbitrary units are different from the other arbitrary units for the rate and other rate coefficients. Importantly, there is a consistent set of measurements and quantities for all experiments, which are directly comparable.
(42) For the experimental data for the two sets of reactions shown in Figures 4 and 5, the values for $[A]_0$ were constrained to a single value for each set of three experimental profiles, and therefore, slightly different values were obtained for curve i in the best fit of the two sets of data.